

AHMAD AYUB NU'MAN

FULL-STACK SOFTWARE ENGINEER

End-to-End Systems Architecture • Full-Stack Development • DevSecOps

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with comprehensive experience directing the end-to-end software development lifecycle (SDLC) from business domain analysis to containerized production deployment. Proven track record architecting institutional education governance (SIMMACI for 270+ institutions) and multi-tenant core financial ERP engines with clean Domain-Driven Architecture (DDD), Segregation of Duties (Maker-Checker), and Money Value Objects. Experienced in rigorous quality assurance (1,800+ automated test cases), high-concurrency database design, and server hardening on Linux/Docker/Coolify.

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript, PHP 8+, SQL, HTML5, CSS3
Frontend Engineering:	React 18/19, Next.js (App Router), Tailwind CSS, Blade, Zustand, Framer Motion, Shadcn UI
Backend & APIs:	Laravel 11/12, NestJS, Node.js, Express, RESTful APIs, Microservices, Role-Based Access Control (RBAC)
Databases & ORM:	PostgreSQL 16, MySQL, Prisma ORM, Schema Design, Composite Indexing, Database Constraints
DevOps & Cloud:	Docker & Docker Compose, Coolify, Vercel, Git / GitHub, Linux/VPS Administration, Nginx (HSTS/CSP), S3 / MinIO
Architecture & Testing:	Domain-Driven Design (DDD), Segregation of Duties (Maker-Checker), Closed-Loop Wallets, State Machines, Automated Testing (PHPUnit, Pest, Jest), DevSecOps

PROFESSIONAL EXPERIENCE

Staff IT & Sistem Informasi (Lead Full-Stack Systems Engineer & Product Owner) 2024 – Present

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- Direct the end-to-end SDLC across institutional software platforms, leading product roadmaps, functional requirements, REST API contracts, and Linux VPS operations.
- Architected and deployed SIMMACI, a decoupled fullstack platform (React 19 SPA + Laravel 12 on PostgreSQL 16) administering 270+ educational institutions, 2,000+ staff (PTK), and 17,000+ students with automated batch decree (SK) generation and student lifecycle synchronization.
- Architected and engineered the Institutional Core Financial System (Aplikasi Keuangan) using clean Domain-Driven Architecture, Money Value Objects (preventing float rounding errors), and multi-tenant TenantScope.
- Engineered critical financial services including JournalPostingService, OpeningBalanceService, FiscalPeriodService, and sequential journal voucher allocation with a strict draft-to-posted state machine and Maker-Checker segregation.
- Executed comprehensive automated testing suites ensuring release stability: 1,822 test cases on SIMMACI (PHPUnit/Pest, 100% PASS, 36,983 assertions) and 63 financial unit/feature tests (193 assertions) verifying balance invariants.
- Hardened infrastructure and application security: remediated SEC-001–004 vulnerabilities, closed 27 API guards, sanitized Git history (1,945+ commits), and tuned Coolify/Docker memory limits with Nginx HSTS & CSP.

SELECTED SOFTWARE ENGINEERING PROJECTS

SIMMACI — Enterprise Education Governance Platform *React 19, TypeScript, Laravel 12, PostgreSQL 16, Tailwind CSS, Vite PWA, PHPUnit/Pest*

Repository: github.com/ayebe51/simmaci

- Architected a decoupled fullstack platform (React 19 SPA + Laravel 12 REST API on PostgreSQL 16) centralizing administration for 270+ educational institutions, 2,000+ staff, and 17,000+ students.
- Implemented automated Word/PDF batch document templating (PHPWord/Docxtemplater), multi-level review queues, S3/MinIO cloud storage isolation, and 1,822 automated PHPUnit/Pest test cases (100% PASS, 36,983 assertions).

Koneksi Santri — Integrated Islamic Boarding School ERP *Next.js 14, TypeScript, NestJS, PostgreSQL, Prisma ORM, Closed-Loop Wallet, Docker*

Live Demo: admin.koneksisantri.tech | Repository: github.com/ayebe51/koneksi-santri

- Architected a closed-loop digital wallet system for cashless on-campus merchant & canteen spending, featuring parent-controlled daily spending caps and instant balance ledgers.
- Engineered an integrated ERP system bridging student administration, guardian communications, automated tuition billing, and multi-branch double-entry general ledger accounting.
- Designed QR-code Smart Gate security checkpoint logging integrated with automated WhatsApp notification dispatches for verified student pickups.

Enterprise POS & Inventory (Kiro ERP) — Retail ERP & Omnichannel POS

React 19, NestJS, TypeScript, Prisma ORM, PostgreSQL, Apache ECharts

Live Demo: inventory-pos-hazel.vercel.app | Repository: github.com/ayebe51/inventory-pos

- Implemented a Domain-Driven Design (DDD) retail management system handling multi-warehouse stock replenishment, procurement 3-way matching, and POS cashier operations.
- Engineered an append-only inventory movement ledger enforcing Weighted Average Cost (WAC) and FIFO valuation models with sub-second transaction consistency.

EDUCATION

Sarjana Hukum (S.H.) — Islamic Economic Law

Graduated 2019

STAI At-Tahdzib Jombang • Jombang, East Java, Indonesia